Mathematical definition of the tortuosity scores

Technical summary

This document is the code-audited reference for every scoring method exposed by the application. The implementation scores **saved vessels**, not the complete skeleton. Geometry is normalized to a common fundus diameter, vessels shorter than the configured threshold are excluded, and the remaining vessel scores are aggregated at eye level.

The six implemented primary methods are Local-bump v1, Local-bump v2, Arc/chord, curvature-squared, Tortuosity Density, and the RDP external-angle sum. Only Local-bump v1 uses an upper-tail component and a separate display multiplier at eye level. All other methods use a simple length-weighted mean.

Common geometry and notation

Let a saved vessel have an ordered centreline

$$P = (\mathbf{p}_0, \ldots, \mathbf{p}_n), \qquad \mathbf{p}_i = (x_i, y_i).$$

The fundus mask area is A pixels. Its equivalent diameter and the coordinate normalization factor are

$$d_{\text{fundus}} = 2\sqrt{\frac{A}{\pi}}, \qquad c = \frac{1024}{d_{\text{fundus}}}, \qquad \mathbf{q}_i = c\,\mathbf{p}_i.$$

If no valid fundus mask is available, the implementation uses $c = 1$. All primary metrics below use the normalized points $\mathbf{q}_i$. The normalized path length and chord are

$$L = \sum_{i=0}^{n-1} \|\mathbf{q}_{i+1} - \mathbf{q}_i\|_2, \qquad C = \|\mathbf{q}_n - \mathbf{q}_0\|_2.$$

By default, a vessel contributes to the eye score only if its geometry is valid and $L \geq 100$ normalized pixels. This filter can be disabled or configured.

Unless stated otherwise, resampling places points every $h = 4$ normalized pixels and includes the final endpoint. A moving average with a default span of five points is used by Local-bump v1, Local-bump v2, and Tortuosity Density.

1. Local-bump v1 (default)

After resampling and smoothing, the segment direction is unwrapped before its first difference is taken:

$$\theta_i = \mathrm{unwrap}(\mathrm{atan2}(y_{i+1} - y_i, x_{i+1} - x_i)), \qquad \Delta\theta_i = \theta_{i+1} - \theta_i.$$

Here, `unwrap` removes artificial jumps at the $-\pi/\pi$ boundary: for example, a direction change from 179° to -179° is treated as 2°, not -358°. It changes only the angle representation, not the vessel geometry.

With the default threshold $\tau = 0.035$ radians,

$$\delta_i = \begin{cases} \Delta\theta_i, & |\Delta\theta_i| \geq \tau, \\ 0, & |\Delta\theta_i| < \tau. \end{cases}$$

If m is the number of angle-difference samples, including samples set to zero, the local-bump energy is

$$E = \frac{1}{m} \sum_{i=1}^{m} |\delta_i|.$$

Let N be the number of sign changes after zero entries have been removed from (δ_i). The oscillation density and vessel score are

$$D = 100 \frac{N}{L}, \qquad B = E\sqrt{D}.$$

A vessel with no sign change has $B = 0$. The vessel-level `primary_score` is the unscaled value B.

For eligible vessels $v \in V$, the global component is

$$G = \frac{\sum_{v \in V} L_v B_v}{\sum_{v \in V} L_v}.$$

For the tail component, vessels are sorted by decreasing B_v. The algorithm takes exactly $\rho \sum_v L_v$ length from that ordering, using a fractional length from the last vessel when necessary. With $a_v \in [0, L_v]$ denoting the length taken from vessel v,

$$\rho = 0.20, \qquad \sum_v a_v = \rho \sum_v L_v, \qquad T = \frac{\sum_v a_v B_v}{\sum_v a_v}.$$

The displayed eye score is

$$S_{\text{eye}} = 1000\,(0.70G + 0.30T).$$

2. Local-bump v2 (experimental)

V2 uses endpoint-safe smoothing, removes the convolution boundary from the angle-difference sequence, and groups consecutive same-sign turns into lobes. Any lobe whose absolute total turn is below $\lambda = 0.15$ radians is removed iteratively; adjacent lobes that then have the same sign are merged.

Let m be the number of angle changes after removal of the filter boundary, let R be the multiset of angle changes belonging to retained lobes, and let K be the number of retained lobes. The implementation defines

$$E_p = \frac{\sum_{\delta \in R} |\delta|}{m}, \qquad N_p = \max(K-1, 0), \qquad D_p = 100\frac{N_p}{L}.$$

The oscillation and angularity components are

$$O = E_p\sqrt{D_p}, \qquad A_{\text{local}} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(\Delta\theta_i)^2}\sqrt{\frac{100}{L}}.$$

With angularity weight $w = 0.25$, the vessel score is already display-scaled:

$$B_{v2} = 1000\left((1-w)O + wA_{\text{local}}\right).$$

The eye score has no upper-tail term and no further multiplier:

$$S_{\text{eye}} = \frac{\sum_{v \in V} L_v B_{v2,v}}{\sum_{v \in V} L_v}.$$

3. Arc/chord

The vessel score is the length of the resolved saved-vessel route divided by the chord between its resolved endpoints:

$$R = \frac{L}{C}.$$

It is invalid when $C = 0$. The eye score is the length-weighted mean only:

$$S_{\text{eye}} = \frac{\sum_{v \in V} L_v R_v}{\sum_{v \in V} L_v}.$$

There is no upper-tail component and no display multiplier.

4. Curvature-squared

The centreline is resampled by default, but it is **not smoothed** in the implemented curvature-squared calculation. Resampling can be disabled. Numerical first and second derivatives are evaluated with respect to cumulative arc length s, and curvature is

$$\kappa(s) = \frac{|x'(s)y''(s) - y'(s)x''(s)|}{\left(x'(s)^2 + y'(s)^2\right)^{3/2}}.$$

The trapezoidal rule approximates the integral, while the denominator uses the original normalized polyline length L:

$$Q = \frac{1}{L} \int \kappa(s)^2 \, ds.$$

The eye score is

$$S_{\text{eye}} = \frac{\sum_{v \in V} L_v Q_v}{\sum_{v \in V} L_v}.$$

There is no upper-tail component and no display multiplier. The unit is inverse normalized-pixel squared.

5. Tortuosity Density

Resampling, smoothing, and thresholding are used only to locate significant changes in curvature sign. Each boundary is placed halfway in arc distance between consecutive non-zero angle changes of opposite sign. The arcs and chords in the formula are then measured on the normalized, unsmoothed input geometry.

If the boundaries create n constant-sign subsegments, with arc length L_i and chord C_i, the vessel score is

$$\mathrm{TD} = \frac{n-1}{L} \sum_{i=1}^{n} \left(\frac{L_i}{C_i} - 1 \right).$$

No significant sign change gives $n = 1$ and therefore $\mathrm{TD} = 0$. A subsegment with zero arc or chord makes the metric invalid. The eye score is

$$S_{\text{eye}} = \frac{\sum_{v \in V} L_v \, \mathrm{TD}_v}{\sum_{v \in V} L_v}.$$

There is no upper-tail component and no display multiplier. The unit is inverse normalized pixels.

6. Sum of external angles after RDP simplification

The normalized centreline is simplified by the Ramer-Douglas-Peucker algorithm with default perpendicular-distance tolerance $\epsilon = 3$ normalized pixels.
For each retained interior bend point, the implementation computes the unwrapped change in segment direction and converts its absolute value to degrees:

$$\phi_i = \frac{180}{\pi} \left| \mathrm{unwrap}(\theta_{i+1}) - \mathrm{unwrap}(\theta_i) \right| .$$

The vessel score is

$$T_{\mathrm{angle}} = \sum_i \phi_i .$$

The eye score is

$$S_{\mathrm{eye}} = \frac{\sum_{v \in V} L_v T_{\mathrm{angle},v}}{\sum_{v \in V} L_v} .$$

There is no upper-tail component and no display multiplier. The result is in degrees.

Cohort-relative comparative score

For every primary method, a report cohort with finite eye scores S_j also receives a min-max comparative score:

$$C_j = \begin{cases} 100 \dfrac{S_j - S_{\min}}{S_{\max} - S_{\min}}, & S_{\max} > S_{\min}, \\ 0, & S_{\max} = S_{\min}. \end{cases}$$

Missing eye scores are filled with the cohort minimum before normalization. This is a cohort rank aid, not an absolute clinical threshold.

Code verification record

Claim	Implementation checked	Automated evidence
Geometry normalization and eligibility	`fundus_coordinate_scale`, `score_saved_vessel`	scaled-geometry and short-vessel tests
Local-bump v1 and upper-tail eye	`local_bump_metrics`, `tail_weighted_mean`,	straight, oscillation, jitter, and tail tests

Claim	Implementation checked	Automated evidence
score	`summarize_eye_score`	
Local-bump v2	`local_bump_v2_metrics`, `persistent_curvature_lobes`	persistent-wave, endpoint, and weak-lobe tests
Arc/chord	`score_segments`, `score_saved_vessel`	central scoring service tests
Curvature-squared	`curvature_squared_metrics`	straight-line, circular-arc, and resampling tests
Tortuosity Density	`tortuosity_density_metrics`	hand-calculated, no-inflection, and invalid-geometry tests
External-angle sum	`rdp_simplify`, `external_angle_sum_metrics`	straight-line and 180-degree staircase tests
Eye aggregation	`scoring.summarize_eye_score`	method-specific aggregation tests
Comparative score	`add_comparative_hybrid_score`, `minmax_series`	formula inspection and full test suite

During this audit, the app/PDF descriptions for Arc/chord and curvature-squared were corrected: both previously mentioned an upper-tail aggregation that the central scoring service does not apply. The curvature-squared description was also corrected to remove a smoothing step that is absent from the calculation.

Limitations and interpretation

- Pixel-derived units become comparable only to the extent that the fundus-mask diameter is a suitable spatial normalization.
- These are deterministic geometric scores; they do not by themselves establish a clinical diagnosis or calibrated risk threshold.
- The comparative score changes when the report cohort changes.
- Local-bump v2 is explicitly experimental and remains separate from the default

Local-bump v1 method.